EFY TESTED
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SMART GLASSES Like GOOGLE GLASS Made With OLED Display

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Google Glass launched a few years ago looks like normal eyeglasses but it is made of a special glass and has a user interface (UI). It works like a smartphone through which you can see time of the day, check news, read books, and much more; the text appears as an overlay or a floating image in the air. At the same time, you can continue seeing your surroundings, just like you do with an ordinary pair of glasses.

Fig. 1: Google Glass

The Google Glass is revolutionary and is likely to become one of the most advanced gadgets in the future. Since these smart glasses are expensive, it is difficult for most people to experience the technology. Hence, we can try and design our own smart glasses.

BILL OF MATERIALS		
Component	**Quantity**	**Description**
Raspberry Pi Zero W	1	SBC for smart glasses
Transparent OLED display	1	2.9cm OLED display
Eye glass	1	Eyeglass

A picture of the Google Glass is shown in Fig. 1 and the author's prototype of smart glasses is shown in Fig. 2. The components required for building the smart glasses, which are somewhat similar to Google Glass, are listed under the Bill of Materials table.

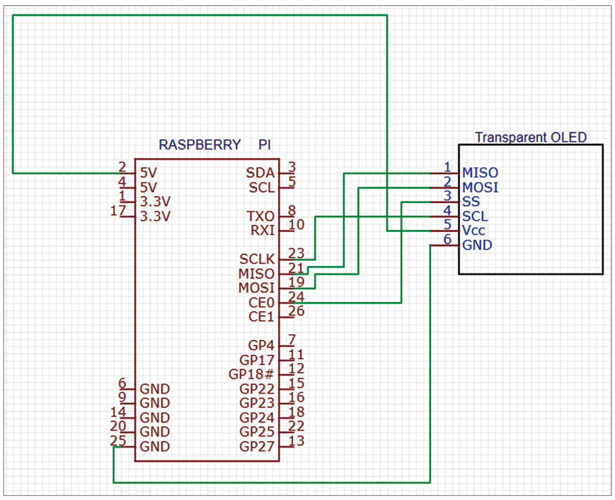

Fig. 3: Circuit diagram for the smart glasses

Fig. 3 shows the circuit diagram of our smart glasses built using Raspberry Pi and a transparent OLED display. Replace a lens or eyeglass in an ordinary pair of glasses with the transparent OLED display and connect it to Raspberry Pi and the battery.

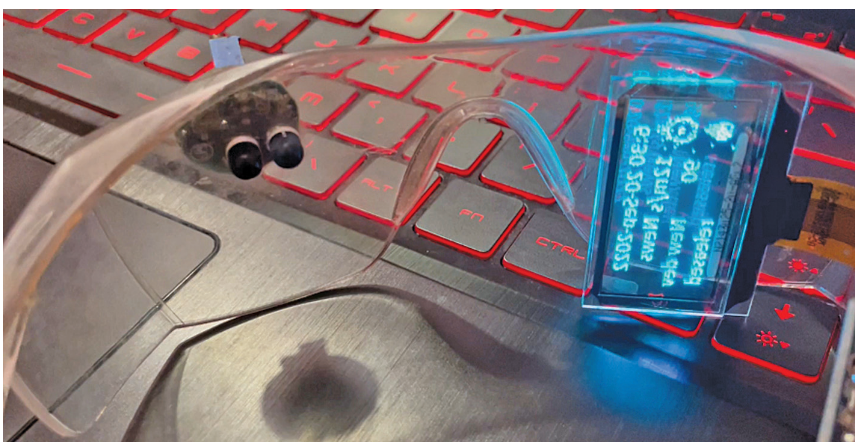

Fig. 2: Author's prototype of the smart glasses

Software

We need Python to create a code for interfacing and configuring the OLED display. The latest version of Raspbian OS comes with Python 3 preinstalled. If you are not using the latest version, install Python and its IDE first. Next, install the libraries and modules for interfacing the display, and then the other Python modules and libraries as per the functions you want for your smart glasses.

A simple reference design of smart glasses that display time and text information on the eyeglasses